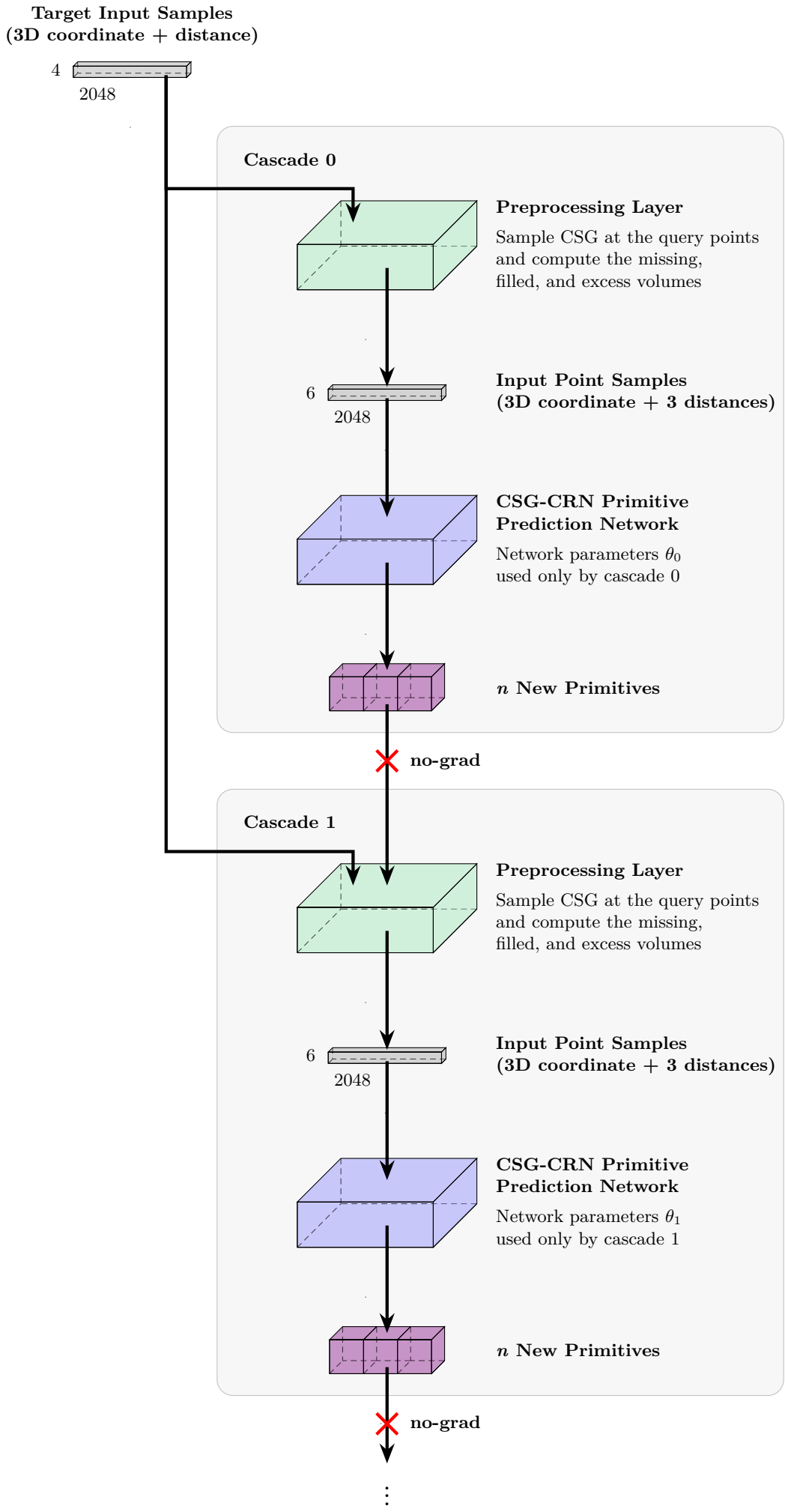

Target Input Samples
(3D coordinate + distance)
4
2048
Cascade 0
Preprocessing Layer
Sample CSG at the query points
and compute the missing,
filled, and excess volumes
6
2048
Input Point Samples
(3D coordinate + 3 distances)
CSG-CRN Primitive
Prediction Network
Network parameters θ_0
used only by cascade 0
n New Primitives
no-grad
Cascade 1
Preprocessing Layer
Sample CSG at the query points
and compute the missing,
filled, and excess volumes
6
2048
Input Point Samples
(3D coordinate + 3 distances)
CSG-CRN Primitive
Prediction Network
Network parameters θ_1
used only by cascade 1
n New Primitives
no-grad